

Philipp Benjamin Aretz

Center for Theoretical Physics, Massachusetts Institute of Technology · Cambridge, MA, USA



RESEARCH INTERESTS

Effective field theory and precision QCD; factorization and resummation for collider observables; medium effects in heavy-ion collisions; mathematical structure of quantum field theory.

EDUCATION

PhD in Physics , Massachusetts Institute of Technology, Cambridge, MA, USA	2024–present
MASt in Mathematics (Part III) , University of Cambridge, Cambridge, UK	2023–2024
Pass with Distinction, 92% — ranked 9th of 294	
BSc in Physics , Heidelberg University, Heidelberg, Germany	2020–2023
1.0 (best possible)	
Thesis: <i>The Keldysh Functional Integral and Bounds on Truncated Expectation Values</i> (advisor: Prof. Manfred Salmhofer)	

RESEARCH POSITIONS

Research Assistant , Heidelberg University, Heidelberg, Germany	2022–2023
CRC/TRR 191 Symplectic Structures in Geometry, Algebra and Dynamics	
Differential delay equations: periodic delay orbits and smooth orbit families. Magnetic billiards: KAM-based extension near convex boundaries.	

PUBLICATIONS AND PREPRINTS

Published

- P. B. Aretz**, M. Salmhofer, *A rigorous Keldysh functional integral for fermions*, J.Statist.Phys. 193 (2026) 28, arXiv:2508.01787 [math-ph].
- P. Albers, **P. Aretz**, I. Seifert, *Families of periodic delay orbits*, J. Differential Equations 410 (2024) 228-250, arXiv:2304.07550 [math].

Preprints

- P. B. Aretz**, K. Lee, S. T. Schindler, I. W. Stewart, *Factorization of elastic, single, and double diffractive pp scattering*, arXiv:2607.07788 [hep-ph].

CONTRIBUTED CONFERENCE TALKS

<i>Diffraction at the LHC</i> , ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna	2026
<i>Generalised Detectors in Heavy Ion Collisions</i> , ESI - New Paradigms for Harnessing Quantum Field Theory at Colliders, Vienna	2026
<i>Diffraction Across Colliders</i> , TASI - Exploring Fundamental Physics from Colliders to the Cosmos, Boulder	2026
<i>Diffraction Across Colliders</i> , SCET 2026, Seoul, South Korea	2026

AWARDS AND FUNDING

Tom Frank Fellowship , Massachusetts Institute of Technology	2024
Jennings Prize , University of Cambridge	2024
DAAD Scholarship , Deutscher Akademischer Austauschdienst	2023

SERVICE AND LEADERSHIP

Treasurer , Edgerton House, MIT	2025–2026
President MIT Rowing Club , MIT Rowing Club	2026–present
Organiser, First Year Seminar , Massachusetts Institute of Technology	2024–2025

TECHNICAL SKILLS

Python (NumPy, pandas, TensorFlow); C++ (ROOT, Pythia); Mathematica; MATLAB; LaTeX.

LANGUAGES

German (native); English (C2, Cambridge Grade A, TOEFL 117/120); French (basic); Spanish (basic); Latin (Latinum).